

Muhammad Hamza Riaz

Full Stack AI Engineer

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Forward-thinking Full Stack AI Engineer with 3+ years of expertise in scalable web applications and high-performance AI systems. Proficient in architecting full-stack SaaS platforms with specialization in advanced RAG pipelines and multimodal LLM applications. Proven track record optimizing enterprise systems and delivering secure, user-focused solutions using Next.js, FastAPI, and PostgreSQL. Passionate about zero-knowledge security and cryptography-driven AI applications.

Technical Skills

Languages: Python, JavaScript, Typescript, Bash, SQL, HTML, CSS, PHP

Frontend: Angular, React, Next.js, Tailwind CSS, Hero UI, Web Workers, GSAP, Framer Motion

Backend: Node.js, Express.js, Django, FastAPI, Flask, Laravel, Prisma, SQLAlchemy, REST APIs, Web Sockets

Databases: MongoDB, PostgreSQL, Redis

AI Engineering: LLM Applications, RAG Pipelines, BM25 Retrieval, Reranking, Cross-Encoders, Vector Search, AI Integrations

DevOps & Tools: Docker, CD/CI, Git/GitHub, Postman, Linux

SEO: Keyword Research, On-Page Optimization, Performance Optimization

Cybersecurity: PGP, Web Crypto API, Network Security, OSINT Experience

Soft Skills: Critical Thinking, Communication, Cultural Awareness

WORK EXPERIENCE

Full Stack AI Engineer | Bave Holdings LLC

June 2025 – Present

- **Implemented** core feature sets for enterprise-grade SaaS applications, focusing on delivering high-impact functionality and improving user workflow efficiency.
- **Designed** high-performance full stack applications using **Next.js** and **Node.js**, ensuring seamless user experiences for enterprise-grade deployments.
- **Designed** advanced AI-powered RAG pipelines with **pgvector** and semantic reranking, achieving high retrieval accuracy across complex vector datasets.
- **Developed** the integration of intelligent automation workflows and LLM orchestration, substantially reducing manual data processing overhead.
- **Streamlined** deployment cycles using **Docker** and **CI/CD** pipelines, significantly increasing deployment frequency while maintaining stable production environments.
- **Mentored** a cross-functional team of 5+ developers on AI best practices and modern frontend architectures, accelerating feature delivery and improving team efficiency.
- **Defined** the technical roadmap and strategic priorities for AI product lines, aligning engineering initiatives with core business objectives and driving product growth.

Backend / Full Stack Developer | Afryvo Analytics

January 2025 – June 2025

- **Developed** secure authentication and user management systems using **FastAPI** and **SQLAlchemy 2.0**, implementing fully asynchronous database logic for high-performance transaction processing.
- **Deployed** a robust JWT-based authentication layer with token rotation and **HttpOnly cookie** security, significantly mitigating common session hijacking risks.
- **Refined** account protection mechanisms with rate-limited OTP workflows, substantially reducing unauthorized login attempts and improving account recovery efficiency.
- **Managed** the design of scalable REST APIs, optimizing for concurrent request handling while maintaining sub-100ms latency.
- **Standardized** clean code practices and **Alembic** migration workflows, minimizing database-related deployment errors across the engineering team.
- **Leveraged** the **uv** package manager to optimize developer environments, significantly reducing local setup times and ensuring dependency parity.

Full Stack Developer | Information Technology Services

June 2023 – December 2024

- **Built** multiple web applications using **PHP**, improving overall system performance and enhancing user retention through UI/UX refinements.
- **Maintained** and enhanced specialized medical publication platforms, managing complex **XML-based data structures** and abstract indexing for academic journals.
- **Implemented** critical feature updates and content management workflows for healthcare-focused web applications, ensuring reliable access to medical research data.

- **Collaborated** on the technical maintenance of legacy systems, optimizing XML parsing logic and ensuring stable performance for high-traffic journal repositories.

PERSONAL PROJECTS

Dream Veil

- **Developed** a privacy-first, AI-powered dream journaling and interpretation platform using **Next.js**, **FastAPI**, and **PostgreSQL 15+**, featuring a zero-knowledge "Blind Vault" **E2EE** architecture via the **Web Crypto API** (AES-256-GCM, AES-KW key wrapping, PBKDF2) with **IndexedDB**-persisted DEKs and **Redux** ephemeral key state.
- **Designed** a production-grade **6-Phase RAG Pipeline** implementing **Hybrid Retrieval** (Dense **pgvector** HNSW + Sparse **Postgres BM25**), **Reciprocal Rank Fusion (RRF)**, local **Cross-Encoder** reranking (**ms-marco-MiniLM**), a cosine-similarity **Grounding Guard**, and tiered chronological context assembly for narrative evolution tracking.
- **Architected** a **Zero-Knowledge RAG** flow where ephemeral plaintext powers vector matching and augmented **gemini-embedding-001** embeddings before being irrecoverably discarded, ensuring the server never persists unencrypted psychological narratives or interpretations.
- **Built** real-time, low-latency AI interpretation streaming via stateless **Server-Sent Events (SSE)** in FastAPI, orchestrating a **4-Phase zero-knowledge handshake** across REST and SSE boundaries with **Redis**-cached RAG context, fast/slow-path hash deduplication, and structured metadata extraction.
- Integrated user-selectable **Gemini 3.5/2.5 Flash** models with configurable thinking depth, **multimodal vision** (up to 5 images), and **BYOK** user-owned API keys, managed by a thread-safe **Redis** key rotation system with tiered TTL exhaustion handling and relaxed Gemini safety thresholds for dream imagery.
- **Designed** a fault-tolerant, **5-Phase resumable migration state machine** with intelligent batch re-scanning, enabling secure bulk encryption/decryption of historical libraries across network interruptions without data corruption or premature key deletion.
- **Implemented** **JWT** and **OAuth** (Google, GitHub) authentication with **SlowAPI** rate limiting, strict **CSRF/CORS** origin validation, and deployed the backend serverlessly to **AWS Lambda** via **Docker**, **Serverless Framework**, and **GitHub Actions CI/CD** with **SQLAlchemy 2.0** async and **Alembic** migrations.

Next PGP

- **Developed** **Next PGP**, a zero-knowledge **PGP cryptographic workstation** PWA using **Next.js**, **Auth.js**, **PostgreSQL (Prisma)**, and **OpenPGP.js** — supporting multi-algorithm key generation (**Curve25519**, NIST, Brainpool, RSA), keyring management, keyserver import/publish, and batch, file, and recursive folder encryption entirely client-side.
- **Designed** a custom **45-byte NP binary vault format** with **AES-256-GCM**, **PBKDF2-SHA512** (1M iterations), **HMAC-SHA256** integrity checks, **DEFLATE** compression, and client-side **verification ciphers** — ensuring the server never receives vault passwords or plaintext keys.
- **Built** a secure backend with **probabilistic rate limiting** (99% cleanup overhead reduction), **HMAC-signed CSRF tokens** with timing-safe validation, server-side **cipher format validation**, and a dual **keyserver proxy** with automated failover.
- **Implemented** layered runtime protection: **app-password** encryption of **IndexedDB** keys via **PBKDF2-SHA256**, in-memory-only vault passwords, **HttpOnly JWT** vault session cookies (30-min expiry), and **Argon2id**-hashed OTP vault deletion via Gmail.
- **Improved** cryptographic performance with a dynamic **Web Worker Pool** sized via `navigator.hardwareConcurrency`, parallelizing **OpenPGP** workloads across all CPU cores to keep the UI responsive during heavy batch and folder operations.
- **Shipped** a cross-platform **PWA** with **Hero UI**, offline service workers, **OAuth** (Google, GitHub, Discord), comprehensive security headers (**HSTS**, **CSP**, **COEP**), and encrypted zero-knowledge cloud backup and sync.

Dimitrescu Wallpapers

- **Developed** a large-scale **416-route Next.js 15** fan gallery **PWA** spanning **298** masonry gallery and **112** model-viewer SCP pages for Bela, Cassandra, and Daniela Dimitrescu from Resident Evil Village.
- **Built** **268** leaf gallery pages with **Masonry** layout, **GLightbox** one-click downloads, and **GitHub API**-driven wallpaper inventories, coordinated by dual preloader systems (**Framer Motion** curtain + legacy **GSAP** progress bar).
- **Created** homepage scroll choreography with **GSAP ScrollTrigger**, **Framer Motion** hero carousel, **Lenis** smooth scroll, and **AOS** page transitions.
- **Implemented** per-route **SEO** on all 416 pages with **OpenGraph**, **Twitter Cards**, **JSON-LD**, and automated **sitemap** generation.
- **Shipped** an **Auth.js** review system (Google, GitHub, Discord) with **Prisma**, **IP-based rate limiting**, and input validation; integrated **Character.ai**, **Wattpad**, **Nexus Mods**, and **YouTube** with **Workbox** PWA caching and **Google Analytics**.

EDUCATION

Bachelor of Science in Information Technology | *Virtual University, Karachi*
Diploma in Software Engineering | *Aptech Pakistan, Karachi*

2024
2023